

Project Proposal

Watermark-Guided Tamper Localization and Recovery: A Key-Seeded Self-Embedding Fragile Watermarking Scheme with Joint Localization and Restoration Assessment

Project work for

B.E. in Computer Engineering

At

Department of Computer Engineering

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1. **Project Group ID : [TO BE FILLED]**
2. **Title of the project : Watermark-Guided Tamper Localization and Recovery: A Key-Seeded Self-Embedding Fragile Watermarking Scheme with Joint Localization and Restoration Assessment**
3. **Domain : Information Security and Digital Image Forensics (Multimedia Security)**
4. **Team Member**

PRN	Roll No	Name	Role/Task Assigned	Email	Mobile	Sign

5. **Sponsorship details, if any (Name, External Guide name and Designation with Signature, e-Mail ID)**

None. This is an in-house academic project carried out within the Department of Computer Engineering, with no external sponsor.

6. **Internal Guide Name (with signature of approval) : [TO BE FILLED]**
7. **Type of Project : Research and development — in-house software project with an experimental evaluation component.**

Abstract (150 words)

Digital images are relied upon as records, yet the formats they are stored in protect nothing about their content. Passive forensic detectors can indicate that a region was altered but cannot restore what was destroyed, and their learned instances need annotated corpora and accelerator hardware. This project will develop the active alternative: a deterministic, training-free self-embedding fragile watermarking system. The image will be partitioned into 8×8 blocks, and the two least-significant bit-planes of each block will carry a 32-bit keyed authentication tag computed over that block's six most-significant bit-planes, together with a 96-bit compressed recovery descriptor of a spatially distant partner block chosen by a key-seeded permutation. Verification will recompute each tag to localize tampering at block granularity, rebuild flagged blocks from surviving partners, and mark the remainder explicitly unrecoverable rather than fabricating content. The system will be evaluated across four tamper classes and three tamper ratios, reporting imperceptibility, localization and recovery together.

Keywords (5 to 10) :

Fragile watermarking; self-embedding; tamper localization; image authentication; content recovery; block mapping; keyed hashing; digital image forensics.

I. Problem Definition

Given a digital image stored in a lossless container and a secret key held by the authenticating party, the system to be built must embed authentication and recovery information into the image itself, so that at any later point three questions can be answered from the received image alone.

First, was the image modified, and if so, exactly where? The modified region must be localized to the block grid without reference to the original image and without any external database or side-channel, because in the applications that motivate this work the original is precisely what is unavailable.

Second, what was there before? The original content of the modified region must be approximately restored from information that survived elsewhere inside the same image,

not generated by inference. A reconstruction that is merely plausible has no evidentiary value.

Third, what cannot be restored? Any region whose backup was itself destroyed must be reported as unrecoverable rather than filled in, so that the output never presents invented content as recovered content.

The embedding must be perceptually transparent, the verification decision must be deterministic and independently recomputable by any third party holding the key, and the whole pipeline must run on a commodity CPU with no trained parameters and no accelerator.

II. Process Diagram, System architecture, List of modules, and functionalities

The system will be built as four decoupled stages. Stage 1 embeds at or near acquisition; Stage 2 is the tampering the harness will simulate; Stage 3 verifies and localizes at any later point; Stage 4 recovers. Only the received image, the key and the image identifier will be required at verification time — no original and no reference database. The block diagram below shows the data flow to be implemented.

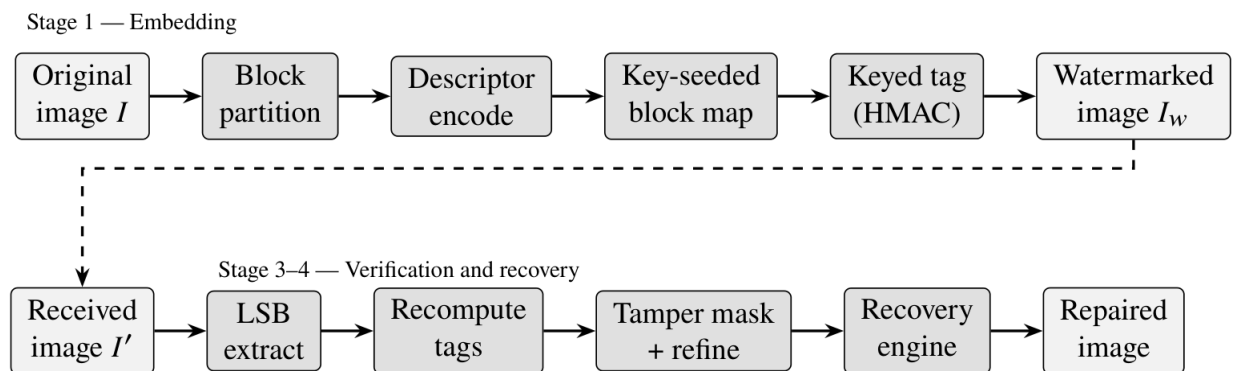


Figure 1: System architecture — the four stages to be implemented, and the data that crosses between them.

Stage 1 — Embedding. The image will be split into 8×8 blocks. For every block the six most-significant bit-planes will be projected out (the MSB view, a & 0xFC) and a 96-bit recovery descriptor computed from it. A key-seeded permutation will assign each block a distant partner, and each block will be written with 128 bits into its two least-significant

bit-planes: its partner's 96-bit descriptor plus a 32-bit truncated keyed HMAC-SHA256 tag binding the block index, the image identifier, the MSB view and the descriptor being carried. Because the tag is computed over the MSB view only, and only LSBs are written, embedding cannot invalidate what it has just signed — so a single pass will suffice, with no iteration to a fixed point.

Stage 3 — Verification and localization. Verification will re-derive the same subkeys from the key, read the stored payload out of the two LSB planes, recompute each block's tag from the block's current MSB view, and compare. Any mismatch flags that block. The flagged set will be refined by a neighbourhood rule and expanded to a pixel mask. Every decision will be recorded in a per-block audit record so a third party holding the key can repeat it.

Stage 4 — Recovery. The flagged set will be partitioned into blocks whose partner survived and blocks whose partner did not. The first group will be rebuilt from the descriptor the surviving partner carries; the second will be marked unrecoverable and left visibly flagged. The recoverability rate $\rho = 1 - |U|/|T|$ will be reported alongside every recovery figure, so that reconstruction quality is never quoted without its coverage.

List of modules and their functionality. The implementation will be split into nine modules, each independently testable:

- a. **Payload foundation** — key derivation with domain-separated subkeys, bit budget, block geometry, MSB projection, keyed tag computation, descriptor encoding and decoding, and LSB packing. Every other module will import it, so the constants exist in exactly one place.
- b. **Block mapper** — builds the key-seeded single-cycle permutation, enforces the minimum spatial separation between a block and its partner, and self-checks bijectivity, absence of fixed points and the separation bound before returning.
- c. **Embedder** — Stage 1. Computes descriptors, permutes them, computes the binding tags and writes both into the two LSB planes, asserting that the six MSB planes are untouched.

- d. **Detector and localizer** — Stage 3. Extracts the stored payload, recomputes tags, produces the block-level tamper mask, applies neighbourhood refinement and expands to a pixel mask.
- e. **Recovery engine** — Stage 4. Splits the flagged set into recoverable and unrecoverable, rebuilds the former from partner descriptors, marks the latter, and reports ρ .
- f. **Tamper harness** — generates the four tamper classes with exact ground-truth masks and the achieved tamper ratio, so that scoring never depends on the detector under test.
- g. **Metrics and evaluation** — PSNR, SSIM, confusion counts, localization scores, in-region recovery metrics, and the CSV result grid.
- h. **Automated gate** — validates the result grid against expected bands and refuses to emit any table or figure if a check fails, so no number can reach the report without passing it.
- i. **Applications** — a local web application and a demonstration interface built over the same pipeline, with no duplicated numerics.

III. Literature survey (Study, Current market survey: similar systems / products available, their pros and cons)

Passive forensics and learned detectors. Passive methods analyse the received image alone. Zhou et al. [13] use a two-stream RGB and noise-residual architecture for splice localization, and ManTra-Net [14] uses self-supervised anomaly features to localize forgeries without training on the specific manipulation. **Pros:** they need no cooperation at capture time and work on arbitrary images. **Cons:** they require annotated corpora and accelerator hardware, their output is a per-pixel confidence rather than an auditable decision, and they cannot restore destroyed content.

Active watermarking. Active schemes insert authentication information before distribution, so verification becomes a decision about self-consistency. Robust watermarks [2] survive processing and serve copyright; fragile watermarks [5], [6] break under any change and serve authentication; semi-fragile schemes [4] sit between. Holliman and Memon [3] showed that block-independent schemes admit counterfeiting by block transplantation, and Fridrich [7] showed that localization granularity trades against security

— both results directly constrain the design proposed here, and are the reason the tag will bind the block index and image identifier as well as the content.

Self-embedding and content recovery. Fridrich and Goljan [1] introduced hiding a compressed version of each region elsewhere in the same image; Lin et al. [8] added a hierarchical structure. **Pros:** the backup travels with the image and needs no database. **Cons:** the backup occupies payload capacity and can itself be destroyed.

The tamper coincidence problem. If a block and its backup fall inside the same tampered region, both are lost. AuSR3 [16] addresses this with a block mapping that guarantees a minimum spatial distance between a block and its partner. This project adopts that idea and credits it; it is not claimed as a contribution.

Reference sharing and fountain-coded allocation. Zhang et al. [9] and Korus and Dziech [10] allocate reference bits across many blocks rather than one-to-one, so recovery degrades gracefully at high tamper ratios. This is stronger than one-to-one mapping under dispersed attack and is the identified upgrade path for this work, stated here rather than concealed.

Learned proactive watermarking. EditGuard [17] embeds a learned watermark that both localizes and recovers. **Pros:** strong reported accuracy. **Cons:** training data, GPU inference and a model artefact to version and defend. No accuracy claim will be made against it; the distinction pursued here is determinism and auditability, not accuracy.

Deployed products fall into four groups, none of which offers localization and recovery together from inside the image itself.

Provenance metadata (C2PA Content Credentials). A cryptographically signed manifest recording an asset’s origin and edit history. **Pros:** an open standard with camera and editing-software backing. **Cons:** the manifest lives beside the pixels, so re-encoding, a screenshot or a metadata strip removes it entirely, and it can never restore content — it attests to history rather than protecting it.

Controlled-capture services (Truepic and similar). Authenticate images at the moment of capture through a proprietary pipeline. **Pros:** strong provenance where deployed. **Cons:** require the vendor’s software and infrastructure, and do not recover destroyed content.

Forensic analysis tools (Amped Authenticate, FotoForensics-style services). Apply passive heuristics — error-level analysis, noise and CFA inconsistency, metadata review — with no prior cooperation. **Pros:** work on arbitrary images. **Cons:** results need expert interpretation, carry no ground truth, and offer no restoration.

Whole-file digital signatures (as standardized for DICOM). **Pros:** mathematically decisive and already deployed in hospital systems. **Cons:** detect only that a file changed — they neither localize the change to a region nor recover it.

Research gap. The gap in the market mirrors the gap in the literature: existing products either attest to provenance externally or analyse pixels after the fact, and none embeds recoverable content inside the image so that a damaged region can be restored from what survived. The gap this project will address is stated narrowly and deliberately. It is not the distant-mapping idea, which is credited to AuSR3 [16], and it is not superiority over learned proactive watermarking [17]. It is threefold: a construction in which every positive detection reduces to one recomputed keyed hash comparison that a third party can repeat bit for bit; an explicit keyed construction with each field binding stated and analysed, including what each binding defeats; and joint evaluation, in which imperceptibility, localization and recovery are reported together with the unrecoverable fraction disclosed rather than excluded.

IV. Objectives

- a. To design a per-block payload that fits entirely within two least-significant bit-planes, so that the embedding distortion is analytically bounded before a single image is processed, and to achieve mean imperceptibility of at least 40 dB PSNR over a standard corpus.
- b. To construct a key-seeded block mapping that is provably bijective, free of fixed points and short cycles, and that enforces a minimum spatial separation between a block and its recovery partner, so that a localized attack cannot destroy both.
- c. To localize region-level tampering at block granularity with block-level recall of 1.0 across four distinct tamper classes and three tamper ratios.
- d. To produce zero false positives on unmodified watermarked images, verified over a null condition of more than one million independent block verifications.

- e. To reconstruct flagged regions from partner-held descriptors, report an explicit recoverability rate, and mark unrecoverable regions rather than fabricating content for them.
- f. To deliver a reproducible evaluation grid and working software, gated by automated checks that must pass before any result is reported.

V. Scope of the project

In scope. Still images, greyscale and three-channel colour, at 8 bits per channel, stored in lossless containers (PNG, BMP, TIFF). Block sizes 8 and 4, so that the resolution against payload trade-off can be measured rather than asserted. Two recovery descriptor designs to be evaluated against each other. Four tamper classes — copy-paste splicing, object removal by inpainting, crop-and-refill, and random block corruption — at three tamper ratios. A fixed 32-image evaluation corpus, every file pinned by SHA-256 so the corpus reproduces byte for byte. A local web application and a demonstration interface over the same pipeline.

Out of scope. Video and audio. Robustness to lossy re-encoding, resizing, rotation or filtering — the scheme is fully fragile by design, and any change to the pixels is intended to break it. Adversaries in possession of the secret key, against whom no active scheme can defend. Real-time embedding inside camera firmware, which is an engineering integration rather than a research question. Comparison against learned detectors on their own accuracy terms.

VI. Software and hardware requirements

- a. **Language.** Python 3.10 or newer. Cryptographic primitives will be taken from the standard library (hashlib, hmac, struct); no third-party cryptography will be introduced.
- b. **Core libraries.** NumPy, OpenCV (headless), scikit-image and Pillow, each pinned to an exact version so that a result can be reproduced later.
- c. **Interface libraries.** Flask for the local web application, Streamlit for the demonstration interface, Matplotlib for figures.
- d. **Storage.** SQLite for the local library of protected images.

- e. **Machine learning.** None. No framework, no trained parameters, and no model artefact to version — this is a design decision, not a limitation.
- f. **Processor.** Any commodity x86-64 or ARM CPU; execution is single-threaded.
- g. **Accelerator.** No GPU at any stage — embedding, verification or recovery.
- h. **Memory and storage.** 8 GB RAM and about 2 GB of disk for the corpus, the result grid and the generated figures.
- i. **Operating system.** Platform independent; development and testing will be on Windows, with no OS-specific dependency.
- j. **Document build (optional).** tectonic and pandoc, required only to rebuild the paper and this proposal.

VII. Expected Outcomes

The deliverable will be a working, reproducible pipeline together with the evidence needed to defend it. The targets below are the criteria against which the project will judge itself.

Imperceptibility. Mean PSNR of at least 40 dB with SSIM above 0.98 over the 32-image corpus, for every descriptor variant.

Localization. Block-level recall of 1.0 across all tamper classes and ratios, with pixel-level precision reported as the informative figure. Block-level precision will deliberately not be headlined: under the ground-truth rule used here a block-level false positive is structurally impossible, so that metric would read 1.000 regardless of detector quality.

False positives. Zero false-positive blocks over a null condition of more than one million independent verifications on unmodified watermarked images, reported as a determinism and implementation-consistency check — not as a statistical bound on the cryptographic false-accept probability, which that experiment does not measure.

Recovery. An explicit recoverability rate ρ reported at every tamper ratio, in-region recovery quality reported separately from whole-image quality, and every unrecoverable region marked rather than filled. We expect in-region quality to stay roughly flat as the tamper ratio grows while whole-image quality falls, which would identify mapping coverage rather than descriptor fidelity as the dominant constraint — and would point directly at reference sharing as the highest-value extension.

Reproducibility. A complete result grid in CSV, generated tables and figures, an automated gate that must pass before any number is quoted, and an end-to-end test that a third party can run from a clean clone.

Dissemination. An IEEE-format manuscript prepared for submission to a peer-reviewed venue in image forensics or multimedia security — the IEEE International Workshop on Information Forensics and Security (WIFS) and the IEEE International Conference on Image Processing (ICIP) are the two candidate venues identified, both appropriate in scope.

VIII. Probable date of completion

[TO BE FILLED] — as per the departmental academic calendar. The work is planned in nine phases:

Ph.	Activity	Deliverable
1	Literature survey and problem formulation	Survey and gap statement
2	Scheme design: bit budget, tag construction, block mapping	Design specification
3	Implementation of the core pipeline	Embed, detect and recover modules
4	Tamper harness and metrics	Four tamper classes with exact masks
5	Full experimental grid and validation	Result grid and automated gate
6	Applications and adversarial testing	Web application and demo interface
7	Paper preparation and internal review	IEEE-format manuscript
8	Reference-sharing extension	Prototype and comparison
9	Final report, demonstration and viva	Report, presentation, submission

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